THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** moves in a straight line in one direction	distance = |displacement|
2	**(d)** 88 m and 0	circumference = 2πr; returns to start
3	**(a)** 18 m	$s = ut + \frac{1}{2}at^2$
4	**(c)** 50 N	W = mg
5	**(a)** increase the time of stopping and reduce the force	smaller acceleration
6	**(a)** Paddle pushes water backwards; water pushes paddle forwards	act on two different bodies
7	**(c)** 30 N	effort × effort arm = load × load arm
8	**(b)** 60 W	P = W/t
9	**(a)** 100 J	W = F × s
10	**(a)** 340 m s^{-1}	$v = \nu \lambda$
11	**(c)** 5 Hz	oscillations ÷ time
12	**(c)** vacuum	needs a medium

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	No; the straight-line change in position cannot exceed the actual path length (2).	2
14	The force that produces an acceleration of 1 m s^{-2} in a body of mass 1 kg (2).	2
15	Whenever one object exerts a force on a second, the second exerts an equal and opposite force on the first (2).	2
16	KE = $\frac{1}{2}mv^2$ (1); PE = mgh (1).	1+1
17	$v = \nu \lambda$ (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Straight line inclined to the time axis (1); take two points, acceleration = $(v_2 - v_1)/(t_2 - t_1)$ = slope (2).	1+2
19	Airbags and seat belts increase stopping time, reducing acceleration and force (1.5); a fielder pulling hands back while catching a cricket ball (1.5).	1½+1½
20	Displacement is zero (1.5); the tiredness is due to the use of internal body energy (1.5).	1½+1½
21	An electric bell in a jar; air is removed by a pump (1); sound becomes fainter and almost inaudible (1); sound needs a medium (1).	1+1+1
22	Density against distance curve (1); highest point crest, lowest trough (1); distance between two crests is the wavelength (1).	1+1+1
	OR: Time gap 0.1 s; total distance 340 × 0.1 = 34 m; to and fro, so 17 m (3).	3
23	Average speed = 400/5 = 80 km h^{-1} (2); average velocity = 0 (1).	2+1
	OR: Slope zero, so acceleration = 0 (1); displacement = area = 20 × 6 = 120 m (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Exhaust gases pushed down; the upward reaction exceeds weight (2). (ii) Engines fire against the direction of motion, slowing it (1). (iii) Air rushes out, balloon moves opposite (2).	2+1+2
	OR: (i) 2 N (1). (ii) 20 m s^{-2} (1). (iii) 0.4 m s^{-2} (1). (iv) Same force but different masses (2).	1+1+1+2
25	(i) Reflected sound heard after the original, gap ≥ 0.1 s (1). (ii) 17 m (1). (iii) Reverberation has gaps < 0.05 s; it is repeated reflections (1). (iv) Absorbing panels, curtains, upholstered seats (2).	1+1+1+2

	OR: (i) Plucked rubber band on a box (2). (ii) Bell jar experiment (2). (iii) No medium in space (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** displacement	v × t
2	**(b)** 40 m	$v^2 = u^2 + 2as$
3	**(b)** along the tangent	tangent to circle
4	**(b)** inertia	tendency to resist change
5	**(d)** increasing the time over which the passenger stops	reduces force
6	**(a)** 30 N	F = ma
7	**(b)** energy	W = ΔE
8	**(b)** 20 N	effort × effort arm = load × load arm
9	**(d)** change the direction of the effort	MA = 1
10	**(b)** 340 m s^{-1}	$v = \nu \lambda$
11	**(d)** 0.2 s	$T = 1/\nu$
12	**(b)** 20 Hz to 20 000 Hz	average healthy ear

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Distance = 2 × 22/7 × 7 = 44 m (1); speed = 44/40 = 1.1 m s^{-1} (1).	1+1
14	The tendency of a body to resist a change in its state of rest or uniform motion (1); it depends on mass (1).	1+1
15	Work done on an object equals the change in its energy (2).	2
16	When force is zero, displacement is zero, or force is perpendicular to displacement (2).	2
17	Compression: region of higher density; rarefaction: region of lower density in a medium (1+1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Uniform: straight line (1.5); non-uniform: curve, since displacements in equal intervals are unequal (1.5).	1½+1½
19	The paddle pushes water backwards (1); water pushes the paddle forwards with an equal force (1); the forces act on different objects, so the canoe moves forward (1).	1+1+1
20	KE = ½mv^2; with 2v it becomes 4 × ½mv^2 (3).	3
21	Stretch a rubber band over a box and pluck it: vibrations are seen and sound is heard (1.5); when the vibration stops, the sound stops (1.5).	1½+1½
22	A girl runs 100 m east and returns 40 m (1); distance = 140 m (1); displacement = 60 m east (1).	1+1+1
	OR: Area of trapezium = ½ × (5 + 15) × 10 = 100 m (3).	3
23	The foot pushes the ground backwards (1); friction from the ground acts forward on the foot (1); without friction we would slip (1).	1+1+1
	OR: Weight = 30 × 9.8 = 294 N (2); she applies an equal upward force of 294 N (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Horizontal line; inclined line (2). (ii) Slope (1). (iii) ½ × (5 + 15) × 10 = 100 m (2).	2+1+2
	OR: (i) Time on x-axis, position on y-axis, scale, plot points and join (2). (ii) Velocity (1). (iii) Straight line vs curve (2).	2+1+2
25	(i) Work = change in energy (1). (ii) W = ½$m(v^2 - u^2)$, KE = ½mv^2 (2). (iii) ½·15000·v^2 = 367500·100 ⇒ v = 70 m s^{-1} (2).	1+2+2
	OR: (i) P = W/t, watt (1). (ii) Work is the energy transferred; power is its rate (2). (iii) W = 18 000 J; P = 900 W (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** m s^{-2}	change in velocity per second
2	**(b)** 44 m and 0	circumference = 2πr; returns to start
3	**(a)** 20 m s^{-1}	72 × 5/18
4	**(d)** balanced	net force is zero
5	**(d)** 1 kg	definition
6	**(a)** 0 N	add if same direction; subtract if opposite
7	**(a)** h only (and g)	v = √(2gh)
8	**(b)** negative	force opposes displacement
9	**(c)** 4	load ÷ effort
10	**(c)** 3 m	λ = v/ν
11	**(a)** reverberation	gap < 0.05 s
12	**(b)** solids	closely packed particles

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	90 × 5/18 = 25 m s^{-1} (2).	2
14	F = ma = 1500 × 2 (1) = 3000 N (1).	1+1
15	Work done when a force of 1 N displaces a body by 1 m in the direction of the force (2).	2
16	The particles of the medium vibrate parallel to the direction of propagation (2).	2
17	λ: distance between two consecutive crests; ν: oscillations per second; T: time for one oscillation (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	a = (v − u)/t (2); hence v = u + at (1).	2+1
19	a = 50/10 = 5 m s^{-2} (1.5); s = ½ × 5 × 16 = 40 m (1.5).	1½+1½
20	mgh = ½mv² (1); v² = 2 × 10 × 2.5 = 50 (1); v ≈ 7.07 m s^{-1} (1).	1+1+1
21	Speed = distance/time = λ/T (2); since ν = 1/T, v = ν λ (1).	2+1
22	Net force = 5 N, a = 5/25 = 0.2 m s^{-2} (2); s = ½ × 0.2 × 4 = 0.4 m (1).	2+1
	OR: No (1); the applied force balances friction, so net force is zero (2).	1+2
23	Force opposes displacement (1); W = 200 × (−0.15) (1) = −30 J (1).	1+1+1
	OR: Force to lift slowly = mg (1); work = mg × h (1); stored as PE = mgh (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) a = 2, F = 3000 N east (1). (ii) 0 (1). (iii) a = −2, F = −3000 N (westwards) (1). (iv) 25 + 50 + 25 = 100 m (2).	1+1+1+2
	OR: (i) Every action has an equal and opposite reaction on the other body (1). (ii) Paddle pushes water back, water pushes paddle forward (2). (iii) They act on different objects (2).	1+2+2
25	(i) 17.2 m (1). (ii) 1.72 cm (1). (iii) Below 20 Hz is infrasonic; above 20 kHz ultrasonic (1). (iv) Elephants hear infrasound; bats hear ultrasound (2).	1+1+1+2
	OR: (i) Piston model: alternate compressions and rarefactions (2). (ii) No; particles oscillate about their positions and only energy moves (1). (iii) Parallel vs perpendicular vibration (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.